

MA 516: Topology

Course Code	MA 516
Course Name	Topology
L-T-P-C	3-1-0-4
Prerequisites	MA 511(Real Analysis)
Intended for	UG/PG
Mutual Exclusion	None
Approval	45th BoA (Feb 2022); 38th Senate (Feb 2023), in the M.Sc Applied Mathematics curriculum

Course Contents

- **Topological Spaces:** open sets, closed sets, neighbourhoods, bases, subbases, limit points, closures, interiors, continuous functions, homeomorphisms. [7 Hours]
- **Examples of topological spaces:** subspace topology, product topology, metric topology, order topology. [5 Hours]
- **Compactness:** compact spaces and its properties, locally compact spaces, one point compactification, paracompactness, Tychonoff theorem. [7 Hours]
- **Countability Axioms:** first countable spaces, second countable spaces, separable spaces, Lindeloff spaces. [4 Hours]
- **Separation Axioms:** Hausdorff, regular and normal spaces, Urysohn's lemma, Urysohn's Metrization theorem, Tietze extension theorem, partition of unity. [6 Hours]
- **Connectedness:** connectedness, path connectedness, connected subspaces of the real line, components and local connectedness. [5 Hours]
- **Quotient topology:** examples of quotient topology: construction of cylinder, cone, suspension, Mobius band, torus, topological groups, orbit spaces. [5 Hours]
- **Algebraic Topology:** homotopy, deformation retract, contractible spaces, path homotopy, fundamental group. [3 Hours]

Text Books

1. G. F. Simmons, **Topology and Modern Analysis**, Tata McGraw-Hill, 2004.
2. A. Hatcher, **Algebraic Topology**, Cambridge University Press, 2002.

References

1. J. Dugundji, **Topology**, McGraw-Hill Inc., 1988.
2. J. R. Munkres, **Topology: A First Course**, Prentice-Hall, 1975.
3. M. A. Armstrong, **Basic topology**, McGraw-Hill Book Co. (UK), Ltd., 1979.